One-loop superconformal components of $\mathcal{N}=4$ SYM in the Costello BV construction: freedoms, transformation laws, invariants

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References. [M]: the manuscript of `N4_Costello_BV_2026-10-27` (labels as in its sources, chapters 2–12). [C07]: K. Costello, arXiv:0706.1533 (Def 4.3.3 gauge fixing axioms (1)–(5); §11 families of gauge fixings; Thm B, Thm C). [CG2]: K. Costello, O. Gwilliam, *Factorization algebras in QFT* vol. 2. [ESW]: arXiv:2002.10517. [BK]: arXiv:2512.07771. [B03c]: N. Beisert, hep-th/0307015. [B03d]: N. Beisert, hep-th/0310252. [BBH]: G. Barnich, F. Brandt, M. Henneaux, hep-th/0002245. [HB]: `Higher_brackets_from_Costello_2026-10-06`, Remark `rem:scheme`. [TW]: `step2/next/twist/twist.tex` (2026-11-05). [GF]: `~/N4_BV_HK2/gf/PROGRESS.txt` (2026-11-05). [SQ]: `sqed_Q1_biletters.tex`. Checks: `checks.py`, `checks.json` (blocks C1–C9). Unit $=\frac{\hbar\,\mathrm{tr}_{\mathfrak{g}}(\mathrm{ad}_X\mathrm{ad}_Y)}{16\pi^2(X,Y)}=-\frac{g^2N}{32\pi^2}$, $L:=\log\frac{3}{2}$.

1 The object

Definition 1.1 (Data)**.** Fixed: the classical equivariant action $S_{\mathfrak{psl}(4|4)}=S+S^{(1)}+S^{(2)}$ [M, Def c2-action]; a gauge-fixing operator Q^{GF}; a regularisation (Definition 6.1); the counterterm convention (singular parts in $\mathrm{span}\{\varepsilon^{-m}\log^k\varepsilon\}_{(m,k)\neq(0,0)}$). A one-loop scheme is a pair $(\hbar F_1,\psi)$: $sF_1=\mathrm{Ob}$ [M, Thm c8-obstruction (ii)], ψ as in [M, Def c8-renorm]. $L=d_{CE}+\Lambda$ is the local representative [M, Def c8-rep, Thm c8-L], $L=s+\hbar L_1+O(\hbar^2)$, $\Lambda_1:=L_1$. The components are [M, Def c8-components]

$$Q_1(O)=\hbar L_1(O)^{(1)}(\xi),\qquad S_1(O)=\hbar L_1(O)^{(1)}(\eta),\qquad D_1(O)=\hbar L_1(O)^{(1)}(\Delta),$$
$$K_1,\ P_1,\ M_1,\ R_1 \text{ the same along } K_b,\ P_b,\ M_\omega,\ R_r;\qquad \hbar L_1(O)^{(0)},\qquad \hbar L_1(O)^{(2)}(x,y).$$

Definition 1.2 (Operator cochains)**.** $\mathfrak{E}:=$ the $C^\bullet(\mathfrak{psl}(4|4))$-linear operators X on local cochains, independent of ε and T, preserving the Weyl weight of [M, Def c8-quant], local to local. $\mathfrak{E}^n$: total degree n. $X=\sum_{k\geq0}X^{(k)}$, $X^{(k)}$ raising the arity by k. $[X,Y]$: graded commutator. $\partial X:=[s,X]$.

Lemma 1.3. *(i)* $\partial^2=0$. *(ii)* $L\circ L=0$ *mod* $\hbar^2\Leftrightarrow\partial L_1=0$. *(iii) For* $U=1+\hbar\mathfrak{u}$, $\mathfrak{u}\in\mathfrak{E}^0$*:* $U^{-1}LU=s+\hbar(L_1+\partial\mathfrak{u})+O(\hbar^2)$. *(iv)* $\psi\in\mathfrak{E}^0$*;* $\mathrm{ad}_\phi:=\{\phi,-\}\in\mathfrak{E}^0$ *for* ϕ *local of degree* -1*;* $\mathrm{ad}_G\in\mathfrak{E}^1$, $\partial\,\mathrm{ad}_G=\pm\mathrm{ad}_{sG}$ *for* G *of degree* 0.

Proof. (i) $[d_{CE},X]$ is $C^\bullet$-linear for $C^\bullet$-linear X; $s^2=0$. (ii) $L\circ L=s^2+\hbar(sL_1+L_1s)+O(\hbar^2)$. (iii) $(1-\hbar\mathfrak{u})(s+\hbar L_1)(1+\hbar\mathfrak{u})=s+\hbar(L_1+s\mathfrak{u}-\mathfrak{u}s)$. (iv) [M, Def c8-renorm, Def c2-dce]; [M, Lemma c12-deriv]. □

Definition 1.4 (Point operators)**.** $\mathcal{V}:=$ degree-0 cohomology of $\Lambda_0^{(0)}=Q+\{I,-\}$ on local functionals with density at the origin, Killing data centred there [M, Def c12-centred]. $\mathcal{V}=\bigoplus_w\mathcal{V}_w$ by classical Weyl weight, $\dim\mathcal{V}_w<\infty$. $\rho_0(x):=$ map induced by $-\Lambda_0^{(1)}(x)$ [M, Def c12-class]; $\rho_0(\Delta)|_{\mathcal{V}_w}=-2w$ [M, Lemma c9-SDelta (i)]. $\mathfrak{l}:=\mathfrak{so}(V)\oplus\mathfrak{so}(W_{\mathbb{R}}^{\perp})\oplus\mathbb{C}\Delta$, $\mathfrak{n}_-:=V\oplus\Sigma$ $(P_b,\ \xi)$, $\mathfrak{n}_+:=\Sigma^*\oplus V$ $(\eta,\ K_b)$; $\rho_0(x)\mathcal{V}_w\subset\mathcal{V}_{w-\lambda_x/2}$. $\mathrm{End}(\mathcal{V})_0$: weight-preserving endomorphisms; $\mathrm{End}(\mathcal{V})_{\mathrm{h}}$: homogeneous ones.

Hypothesis 1.5. (H0) $L_1^{(0)}$ vanishes on a set of representatives of $\mathcal{V}$.

Remark 1.6. (H0) is reached by ψ when $[\Lambda_1^{(0)}]:\mathcal{V}\to H^1(\Lambda_0^{(0)})$ vanishes [M, Def c10-hyp (ii)]; on local functions at a point $H^1(\Lambda_0^{(0)})=0$ for semisimple gauge algebra (ghost classes are $\mathrm{tr}\,c^{2r-1}$, $r\geq2$ [BBH]).

Definition 1.7 (Induced cocycle)**.** Under (H0), $\delta(x):=$ map induced by $-\hbar\Lambda_1^{(1)}(x)$ on $\mathcal{V}$. Then $\rho_0+\delta$ is a representation of $\mathfrak{psl}(4|4)$ on $\mathcal{V}\otimes\mathbb{C}[\hbar]/\hbar^2$:

$$\delta\in Z^1\big(\mathfrak{psl}(4|4);\mathrm{End}(\mathcal{V})_{\mathrm{h}}\big),\qquad \delta([x,y])=[\delta(x),\rho_0(y)]+[\rho_0(x),\delta(y)]$$

[M, Cor c8-homotopy, Thm c12-opalg (iii)]. Anomalous dimension: $\rho_0(\Delta)+\delta(\Delta)=-2(w+\gamma)$, i.e. $\delta(\Delta)=-D_1$, $D_1=2\gamma$ on eigenvectors.

2 The freedoms

Theorem 2.1 (Complete list)**.** *The inputs of L_1 beyond $S_{\mathfrak{psl}(4|4)}$ are exactly:*

(a) $\hbar F_1 \in F^{\mathrm{ref}} + Z^0$, $Z^0 := \{G$ *local cochain,* $\deg 0$*, weight* 4*, quantum numbers of [M, Thm c12-bound],* $sG = 0\}$*. Inside* Z^0*:*

(a1) $B^0 = s(\Phi^{-1})$*,* Φ^{-1} *:= local cochains of degree* -1*, weight* 4*;* $\phi = \phi^{(0)} + \phi^{(1)} + \phi^{(2)} + \cdots$*.* $\phi^{(0)}$*: field redefinitions, among them the field countings* $\mathcal{N}_f$*,* $f \in \{A|_V, A|_{W_{\mathbb{R}}^\perp}, \lambda, c\}$ *(quadratic parts* $Q\mathcal{N}_f$*: the* Q*-closed directions* $\beta_1, \dots, \beta_4$ *of* $\hbar F_1^{(0)}|_2$*), the inner direction* $S^{(1)}(\Delta)$*, and* $S^{(1)}(y)$*,* $y \in \mathfrak{so}(V) \oplus \mathfrak{so}(W_{\mathbb{R}}^\perp)$ *(non-covariant).* $\phi^{(1)}(x)$*:* x*-dependent redefinitions;* $(Q + \{I, -\})\phi^{(1)}(x)$ *are the* Q*-exact directions of* $\hbar F_1^{(1)}(x)$ *(*κ_V, κ_λ *along* Δ *and their parts along* P_b*).* $\phi^{(2)}(x, y)$*: exact directions of* $\hbar F_1^{(2)}$*.*

(a2) $Z^0/B^0 = H^0(s) \supseteq \mathbb{C}[S - S^{(2)}] \oplus \mathbb{C}[G_B]$ *[M, Thm c8-obstruction (iv), Prop c8-B].*

(a3) Closed directions of $\hbar F_1^{(1)}(x)$ *at fixed* $\hbar F_1^{(0)}$*:* $H^{-1}(Q + \{I, -\})$ *classes (uniform* $t_x S^{(1)}(x)$*; on two legs the sector parts* $S^{(1)}(x)_d$*; along* K_b *also* $G_1^K, \dots, G_4^K$*). The arity-two equations restrict* t *to a diagonal derivation,* $t \in \mathbb{C}\lambda \oplus \mathbb{C}B$ *(C1a: the diagonal derivations of* $\mathfrak{psl}(4|4)$ *on the nine blocks form a* 2*-dimensional space spanned by* λ *and* B*).*

(b) $\psi^{(0)}$*: finite parts of every local operator, gauge-variant ones included: single letters, ghost composites* $c \cdot X$*, antifield composites, composites of any length; non-derivation (second order on words).*

(c) $Q^{GF} \in \mathcal{G}$*,* $\mathcal{G}$ *:= operators satisfying [C07, Def 4.3.3] (1)–(5), covariant under translations,* $\mathfrak{so}(V)$*,* $\mathfrak{so}(W_{\mathbb{R}}^\perp)$*, scaling; only path components of* $\mathcal{G}$ *enter (Theorem 2.4).*

(d) The regularisation: profile $\varpi \in \{\varpi$ *smooth, rapidly decreasing on* $[0, \infty)$*,* $\varpi(0) = 1\}$ *(Definition 6.1), the unit of length in* $\log \varepsilon$*, or any other regularisation of the one-loop graphs (point splitting, Pauli–Villars, DR, BMHV, DRED) completed by finite local counterterms to the equivariant QME.*

(e) Labels: (e1) tree-level automorphisms U_0 *of local cochains with* $[U_0, s] = 0$*; (e2) the normalisation of* $\hbar$ *against the unit; (e3) relative normalisation of operators of different length.*

(f) $\psi^{(k)}$*,* $k \geq 1$*: arity-raising components of* ψ *(e.g. a finite part of* $c \cdot X$ *with a component linear in* ξ*), the homotopies of the* L_∞ *morphism* $\mathfrak{psl}(4|4) \to \mathrm{End}$ *within its class.*

Proof. [M, Def c8-quant, c8-renorm, c8-rep] use $S_{\mathfrak{psl}(4|4)}$, Q^{GF}, the regularisation, the counterterm convention, $F_{\mathfrak{psl}(4|4)}$, ψ, the unit and the classical labels; L is T-independent [M, Thm c8-L (ii)]. Solutions of $sF_1 = \mathrm{Ob}$: affine over Z^0 [M, Thm c8-obstruction (iii)]. A change of the counterterm convention or of the unit of ε is a finite local counterterm [C07, Thm C], included in (a), (b) by Theorem 2.3. (a3): [M, Thm c12-eq1 (iii), Thm c12-fix (ii)]; C1a. □

Lemma 2.2 (Locality of scheme differences)**.** *Let* R, R' *be two regularisations of the one-loop graphs with the same vertices and the same limit propagator* $P(0, T)$*, each followed by subtraction of local singular parts. For every one-loop graph* G *of superficial degree* D *at generic external momenta* p*,*

$$G_{R'}^{\mathrm{ren}}(p) - G_R^{\mathrm{ren}}(p) \;\in\; \textit{polynomials of degree} \leq D \textit{ in } p \quad (\textit{tree flow of a local functional}).$$

Proof. ∂_p^{D+1} annihilates the subtracted parts (polynomials of degree $\leq D$, [C07, Thm B], [M, Lemma c4-local]); $\partial_p^{D+1} G_R \to \partial_p^{D+1} G$, an absolutely convergent integral, for every R (dominated convergence for multiplicative edge regulators $0 \leq \varpi \leq 1$; continuity in d at $d = 4$ for DR). One loop: no subdivergence. □

Theorem 2.3 (Regularisations)**.** *Let* R, R' *be as in Lemma 2.2, with the same valid* Q^{GF}*, each with the equivariant QME restored by finite local counterterms and local operator counterterms. There are a local cochain* Γ *of degree* 0 *and* $\psi_{RR'} \in \mathfrak{E}^0$ *with*

$$\mathrm{Ob}^{R'} = \mathrm{Ob}^R + s\Gamma, \qquad F^{R'} := F^R + \Gamma, \qquad L_1^{R'}\big(F^R + \Gamma, \psi + \psi_{RR'}\big) = L_1^R(F^R, \psi).$$

Hence the set of one-loop structures L_1 *is the same for* R *and* R'*.*

Proof. Lemma 2.2 on the graphs of the action: $W_{R'}^{\mathrm{ren}} = W_R^{\mathrm{ren}} + \mathrm{tree}(\Gamma)$; the order-$\hbar$ QME defect of $\mathrm{tree}(\Gamma)$ is $\mathrm{tree}(s\Gamma)$; injectivity of the tree flow [M, Thm c8-L (i)]. On graphs with one vertex δO: $O[T]_{R'} = O[T]_R + \mathrm{tree}(\psi_{RR'}(O))$ plus the insertion of Γ; L is fixed by $d_T O[T] = L(O)[T]$ and is unique. □

Theorem 2.4 (Gauge fixing). *Let Q^{GF}_τ, $\tau \in [0,1]$, be a smooth family satisfying [C07, Def 4.3.3] (1)–(5), covariant under translations, $\mathfrak{so}(V)$, $\mathfrak{so}(W^\perp_{\mathbb{R}})$ and scaling. Assume (Hc): the one-loop equivariant QME and local representatives exist for the family over $\Omega^\bullet([0,1])$ [C07, §11]. Then*

$$L_1(1) = L_1(0) + \partial\mathfrak{u} - \textstyle\sum_i a_i \operatorname{ad}_{G_i}, \qquad \mathfrak{u} = \int_0^1 \chi_1(\tau)\, d\tau \in \mathfrak{E}^0, \quad [G_i] \in H^0(s).$$

Proof. Family representative $\tilde{L} = L(\tau) + d\tau\, \chi(\tau)$; $(d_\tau + \tilde{L})^2 = 0$ gives $\partial_\tau L = [L, \chi]$. s is independent of τ ($S_{\mathfrak{psl}(4|4)}$ does not contain Q^{GF}), so $\chi = \hbar\chi_1 + O(\hbar^2)$ and $\partial_\tau L_1 = \partial\chi_1$. The family solution at $\tau = 1$ differs from the chosen $F_1(1)$ by Z^0: Lemma 1.3 (iii), (iv). □

Theorem 2.5 (Reduction). *Every change in (a)–(f) acts as*

$$L_1 \;\longmapsto\; U_0^{-1}\Big(L_1 + \partial\mathfrak{u} - \textstyle\sum_i a_i \operatorname{ad}_{G_i}\Big)U_0, \qquad \mathfrak{u} \in \mathfrak{E}^0,\ [G_i] \in H^0(s),\ U_0 \text{ of (e1)},$$

with $\mathfrak{u} = \operatorname{ad}_\phi$ for (a1), $\mathfrak{u} = \psi'$ for (b), (f), $\mathfrak{u}$ of Theorem 2.4 for (c), $(\Gamma, \psi_{RR'})$ of Theorem 2.3 for (d); (e2) rescales L_1.

Proof. [M, Prop c8-laws (a)–(c)]; Theorems 2.3, 2.4. □

Corollary 2.6 (Homotopy class). *The class*

$$[L_1] \;\in\; H^1(\mathfrak{E}, \partial) \big/ \operatorname{span}\{[\operatorname{ad}_G] : [G] \in H^0(s)\}$$

is invariant under (a)–(d), (f), and transforms under (e1) by U_0. It is the L_∞-homotopy class of the one-loop deformation of the classical equivariant structure s on local cochains: the cocycle condition is $L \circ L = 0$ mod $\hbar^2$ (Lemma 1.3 (ii)), the coboundaries are the conjugations $1 + \hbar\mathfrak{u}$ (Lemma 1.3 (iii)).

3 Transformation laws

Theorem 3.1 (Chain level). *For $\mathfrak{u} = \sum_k \mathfrak{u}^{(k)} \in \mathfrak{E}^0$, the change $\partial\mathfrak{u}$ has the arity components*

$$\begin{aligned}
(\partial\mathfrak{u})^{(0)} &= [\Lambda_0^{(0)}, \mathfrak{u}^{(0)}],\\
(\partial\mathfrak{u})^{(1)} &= [\Lambda_0^{(0)}, \mathfrak{u}^{(1)}] + [\Lambda_0^{(1)}, \mathfrak{u}^{(0)}],\\
(\partial\mathfrak{u})^{(2)} &= [\Lambda_0^{(0)}, \mathfrak{u}^{(2)}] + [\Lambda_0^{(1)}, \mathfrak{u}^{(1)}] + [\Lambda_0^{(2)}, \mathfrak{u}^{(0)}] + [d_{CE}, \mathfrak{u}^{(1)}],
\end{aligned}$$

components by the sign rule of [M, Thm c8-L (iv)]; $[d_{CE}, \mathfrak{u}^{(0)}] = 0$. For $\mathfrak{u} = \operatorname{ad}_\phi$: $\partial\operatorname{ad}_\phi = -\operatorname{ad}_{s\phi}$ with $(s\phi)^{(0)} = \Lambda_0^{(0)}\phi^{(0)}$, $(s\phi)^{(1)}(x) = \Lambda_0^{(0)}\phi^{(1)}(x) + \{\phi^{(0)}, S^{(1)}(x)\}$, $(s\phi)^{(2)}$ by [M, Lemma c12-wz (i)]. For $G = S - S^{(2)}$: $\big(-a\Lambda_0^{(0)},\, 0,\, +a\operatorname{ad}_{S^{(2)}}\big)$. For $G = G_B$: $\big(0,\, -a\, b_x \operatorname{ad}_{S^{(1)}(x)},\, -a\operatorname{ad}_{S^{(2)}(B\cdot,\cdot)+S^{(2)}(\cdot,B\cdot)}\big)$, b_x the B-eigenvalue. Per generator, in arity one:

freedom	*change of $\hbar L_1(\,\cdot\,)^{(1)}(x)$*	*on $\mathcal{V}$*
$\psi' = \mathfrak{u}^{(0)}$ *(b)*	$\hbar[\Lambda_0^{(1)}(x), \psi']$	$[\rho_0(x), \mathsf{e}]$
$\mathfrak{u}^{(1)}$ *(f)*	$\hbar[\Lambda_0^{(0)}, \mathfrak{u}^{(1)}(x)]$	0
$\phi^{(0)}$ *(a1)*	$-\hbar\operatorname{ad}_{\{\phi^{(0)}, S^{(1)}(x)\}}$	$[\rho_0(x), \mathsf{e}]$
$\phi^{(1)}$ *(a1)*	$-\hbar\operatorname{ad}_{\Lambda_0^{(0)}\phi^{(1)}(x)}$	0
$t = c\lambda + aB$ *(a3)*	$-\hbar\, t_x \operatorname{ad}_{S^{(1)}(x)}$	$c\,[\rho_0(x), \rho_0(\Delta)] = -c\lambda_x\rho_0(x)$ *(inner)*, $-a\,\rho_0(Bx)$
$S - S^{(2)}$ *(a2)*	0	0
G_B *(a2)*	$-a\hbar\, b_x \operatorname{ad}_{S^{(1)}(x)}$	$-a\,\rho_0(Bx)$

$b_x = +1$ on $S_+ \otimes W \subset \Sigma$, $S_- \otimes W \subset \Sigma^*$; -1 on $S_- \otimes W^* \subset \Sigma$, $S_+ \otimes W^* \subset \Sigma^*$; 0 on even x. Along Q, S, D, K, P, M, R: $x = \xi, \eta, \Delta, K_b, P_b, M_\omega, R_r$.

Proof. $s = d_{CE} + \Lambda_0^{(0)} + \Lambda_0^{(1)} + \Lambda_0^{(2)}$, $\Lambda_0^{(k)}$ and $\mathfrak{u}^{(k)}$ raising the arity by k, d_{CE} by 1; collect $[s, \mathfrak{u}]$ by arity. $\mathfrak{u}^{(0)}$ acts on the $\mathfrak{psl}$-free factor only. [M, Prop c8-laws, Lemma c12-wz]. The cohomology column: Corollary 3.2; inner: $\{S^{(1)}(\Delta), S^{(1)}(x)\} \equiv \lambda_x S^{(1)}(x)$ mod exact [M, Thm c12-fix (ii) proof]. □

Corollary 3.2 (Cohomology). *Assume (H0) before and after the change. Then*

$$\delta \;\longmapsto\; \delta + [\rho_0, \mathsf{e}] - a\,\rho_0 \circ B, \qquad \mathsf{e} \in \operatorname{End}(\mathcal{V})_0,$$

the coupling direction and all arity-two changes act as 0 on δ, and $\Lambda_1^{(0)}$ changes by $\Lambda_0^{(0)}$-exact terms on cocycles.

Proof. $U^{-1}LU$: arity zero $(U^{(0)})^{-1}\Lambda^{(0)}U^{(0)}$; arity one on a cocycle o: $\Lambda^{(1)} + \hbar([\Lambda_0^{(1)}, \mathfrak{u}^{(0)}] + \Lambda_0^{(0)}\mathfrak{u}^{(1)} \pm \mathfrak{u}^{(1)}\Lambda_0^{(0)})$, the last two exact resp. 0 on o. Hence $(U^{(0)})^{-1}$ intertwines the induced representations; with lifts ι, ι' of $\mathcal{V}$ satisfying (H0), $1 + \hbar\mathsf{e} := \iota'^{-1}(U^{(0)})^{-1}\iota$ mod $\hbar^2$, weight-preserving since $\mathfrak{u}$ is. G_B, coupling: Theorem 3.1. □

4 Invariants of the action on BRST cohomology

Theorem 4.1 (D_1 is canonical)**.** *Assume (H0) and (Hs): $H^0(s) = \mathbb{C}[S - S^{(2)}] \oplus \mathbb{C}[G_B]$ at weight 4 (classical symmetry condition [M, Def c12-cs] and rigidity of the couplings [M, Prop c12-couplings]). Then $\delta(\Delta) \in \mathrm{End}(\mathcal{V})_0$, equivalently D_1 at the origin, is independent of every choice in (a)–(f), of valid gauge fixing (under (Hc)) and of the regularisation; $[\delta(\Delta), \rho_0(x)] = 0$ for all $x \in \mathfrak{psl}(4|4)$ (R7). On an isotypic component $V \otimes \mathbb{C}^m$, V an indecomposable summand with $\mathrm{End}_{\mathfrak{psl}(4|4)}(V) = \mathbb{C}$, $\delta(\Delta) = \mathrm{Id}_V \otimes \mathsf{M}$: the one-loop mixing matrix M, its eigenvectors and eigenvalues are canonical.*

Proof. Corollary 3.2: $[\rho_0(\Delta), \mathsf{e}] = 0$ (e weight-preserving, $\rho_0(\Delta)$ scalar on $\mathcal{V}_w$); $B\Delta = 0$; Theorems 2.3, 2.4. (R7): the cocycle identity at (Δ, x), $\lambda_x\delta(x) = [\delta(\Delta), \rho_0(x)] + [\rho_0(\Delta), \delta(x)]$, with $[\rho_0(\Delta), \delta(x)] = \lambda_x\delta(x)$ by homogeneity. Schur. □

Corollary 4.2. *Off the origin, D_1 on $\int \omega f$ at x_O has the position part $2x^\mu_O P_1|_{e_\mu}$ [M, Thm c12-pos], which depends on the scheme through P_1.*

Proposition 4.3 (The outer cocycle)**.** *(i) $\rho_0 \circ B \in Z^1(\mathfrak{psl}(4|4); \mathrm{End}(\mathcal{V})_{\mathrm{h}})$. (ii) On every direct summand $V \subset \mathcal{V}$ that is simple and generated by its lowest-weight $\mathfrak{l}$-component V_{top} ($\mathfrak{n}_+ V_{\mathrm{top}} = 0$), $\rho_0 \circ B|_V = [\mathsf{b}_V, \rho_0]$, $\mathsf{b}_V \in \mathrm{End}_{\mathfrak{l}}(V)_0$ the B-grading of V. (iii) On a non-split extension $0 \to N \to E \to C \to 0$ of such modules with $\dim \mathrm{Ext}^1_{(\mathfrak{psl}(4|4),\mathfrak{l})}(C, N) = 1$, $\rho_0 \circ B|_E$ is a coboundary.*

Proof. (i) B is a derivation. (ii) $\theta_t := e^{tB}$ fixes the even part pointwise and preserves Σ, Σ^*; $V^{\theta_t} := (V, \rho_0 \circ \theta_t)$ is simple with the same top, so $V^{\theta_t} \cong V$ by a unique Φ_t with $\Phi_t|_{V_{\mathrm{top}}} = 1$; uniqueness gives $\Phi_{s+t} = \Phi_s\Phi_t$, $\Phi_t = e^{t\mathsf{b}_V}$; $\rho_0(\theta_t x)\Phi_t = \Phi_t\rho_0(x)$ at $t = 0$: $\rho_0(Bx) = [\mathsf{b}_V, \rho_0(x)]$. (iii) θ_t acts on the line Ext^1 by a character e^{tb}; $\Phi^E_t := \Phi^C_t \oplus e^{-tb}\Phi^N_t$ in an $\mathfrak{l}$-splitting intertwines. □

Remark 4.4. Proposition 4.3 corrects [M, Thm c12-opalg (iv)] ("a cocycle that on $\mathfrak{psl}(4|4)$ is not a coboundary"): B outer in $\mathfrak{psl}(4|4)$ does not make $\rho_0 \circ B$ non-trivial in $H^1(\mathfrak{psl}(4|4); \mathrm{End}\mathcal{V})$; on simple summands G_B acts by a coboundary.

Theorem 4.5 (The class)**.** *Under (H0), (Hs), (Hc): $[\delta] \in H^1(\mathfrak{psl}(4|4); \mathrm{End}(\mathcal{V})_{\mathrm{h}})/\mathbb{C}[\rho_0 \circ B]$ is invariant under (a)–(d), (f); (e1) acts by conjugation, (e2) by rescaling. Equivalently: the $\mathbb{C}[\hbar]/\hbar^2$-module $\mathcal{V} \otimes \mathbb{C}[\hbar]/\hbar^2$ with $\rho_0 + \delta$, up to isomorphism reducing to 1 mod $\hbar$ and up to the twist $e^{-a\hbar B}$, is invariant. On the parts of Proposition 4.3 (ii), (iii) the quotient by $\mathbb{C}[\rho_0 \circ B]$ is not needed.*

Proof. Corollary 3.2; Proposition 4.3. □

Theorem 4.6 (Blocks of different anomalous dimension)**.** *Let $\delta(\Delta)$ be semisimple, $\mathcal{V} = \bigoplus_\gamma \mathcal{V}^{(\gamma)}$ its eigenspaces (submodules for ρ_0 by (R7)). If the structure exists modulo $\hbar^3$ with δ_2 homogeneous, then for $\gamma \neq \gamma'$*

$$\delta_{\gamma'\gamma}(x) = \big[\rho_0(x),\ (\gamma' - \gamma)^{-1}\delta_2(\Delta)_{\gamma'\gamma}\big],$$

a coboundary: the invariant content of δ is block-diagonal.

Proof. Order $\hbar^2$ of $[\rho(\Delta), \rho(x)] = \lambda_x\rho(x)$ with homogeneous δ_2: $[\delta(\Delta), \delta(x)] + [\delta_2(\Delta), \rho_0(x)] = 0$; block (γ', γ): $(\gamma' - \gamma)\delta_{\gamma'\gamma}(x) = [\rho_0(x), \delta_2(\Delta)_{\gamma'\gamma}]$. □

Theorem 4.7 (Rigidity of simple summands; protection)**.** *Let $V \subset \mathcal{V}$ be a ρ_0-direct summand, simple, generated by V_{top} with $\mathfrak{n}_+ V_{\mathrm{top}} = 0$. Under (H0):*

(i) $\delta(\Delta)|_V = -2\gamma\,\mathrm{Id}_V$.

(ii) If $\gamma = 0$, then $\delta_{VV} = [\rho_0, \mathsf{e}]$, $\mathsf{e} \in \mathrm{End}(V)_0$, and $\mathsf{e} \in \mathrm{End}_{\mathfrak{l}}(V)_0$ when $\delta|_{\mathfrak{so}\oplus\mathfrak{so}} = 0$: every one-loop component on V is removed by one finite part.

(iii) If $\xi \in \Sigma$ and a highest vector $v_0 \in V_{\mathrm{top}}$ satisfy $\rho_0(\xi)v_0 = 0$, and the $\mathfrak{l}$-type of $\xi \otimes v_0$ does not occur in $V_{w_0+1/2}$, then $\gamma = 0$.

Proof. Whitehead's lemma for $\mathfrak{so}(V) \oplus \mathfrak{so}(W^\perp_{\mathbb{R}})$ on each $\mathrm{End}(\mathcal{V}_w)$ gives $\mathsf{e}_1 \in \mathrm{End}(\mathcal{V})_0$ with $\delta|_{\mathfrak{so}\oplus\mathfrak{so}} = [\rho_0, \mathsf{e}_1]$; $\delta' := \delta - [\rho_0, \mathsf{e}_1]$ vanishes on $\mathfrak{so}\oplus\mathfrak{so}$, $\delta'(\Delta) = \delta(\Delta)$, and is $\mathfrak{l}$-equivariant (cocycle identity with $y \in \mathfrak{l}$, homogeneity). (i) Schur and (R7). (ii) $\gamma = 0$: δ'_{VV} is a relative cocycle; $E := V \oplus V$, $\rho_E = \begin{pmatrix} \rho_0 & \delta' \\ 0 & \rho_0 \end{pmatrix}$ is a module, $\mathfrak{l}$-split. $W := 0 \oplus V_{\mathrm{top}}$ is killed by $\mathfrak{n}_+$ (no lower weights), $E' := U(\mathfrak{n}_-)W$ is a submodule mapping onto $E/(V \oplus 0) \cong V$. If $E' \supset V \oplus 0$ then $E' = E$ and $E_{w_0} = W$, false ($E_{w_0} = V_{\mathrm{top}} \oplus V_{\mathrm{top}}$). So $E' \cap (V \oplus 0) = 0$, $E' = \{(\mathsf{e}'v, v)\}$ with e' $\mathfrak{l}$-equivariant, weight-preserving, and ρ_E-invariance gives $\delta' = -[\rho_0, \mathsf{e}']$. (iii) Cocycle identity at $(\xi, \bar\xi)$, $[\xi, \bar\xi] = \frac{1}{2}(\xi, \bar\xi)\Delta + M + R$ [M, Prop c2-special], on v_0: $\delta'(\bar\xi)v_0 \in V_{w_0-1/2} = 0$, $\rho_0(\xi)v_0 = 0$, $\delta'(\xi)v_0 = 0$ by $\mathfrak{l}$-equivariance and the multiplicity condition, $\rho_0(\bar\xi)v_0 = 0$; hence $\frac{1}{2}(\xi, \bar\xi)(-2\gamma)v_0 = 0$, $(\xi, \bar\xi) > 0$ [M, Lemma c8-cc]. □

Corollary 4.8 (The traceless bilinears)**.** *Assume the classical module generated by* $\int \omega(\iota_{w_1}A, \iota_{w_2}A)$, $(w_1, w_2) = 0$, *is a simple direct summand* $V_{20'}$ *of* $\mathcal{V}$ *(half-BPS* $[0,2,0]$*). The null components of* $Q^i_\alpha \otimes [0,2,0] = (\frac{1}{2}, 0) \otimes ([1,2,0] \oplus [0,1,1])$, $60 + 20 = 80$, *and of* $\bar{Q}_{\dot{\alpha}i} \otimes [0,2,0] = (0, \frac{1}{2}) \otimes ([0,2,1] \oplus [1,1,0])$ *are* $[1,2,0]$, $[0,2,1]$, *absent from* $V_{5/2}$. *Hence* $\gamma = 0$ *and* $\delta|_{V_{20'}} = [\rho_0, \mathsf{e}]$*:* $Q_1, S_1, P_1, K_1, M_1, R_1, D_1$ *on* $V_{20'}$ *are one coboundary. On* $V_{20'}$, *with* $\mathsf{e} = f_X$ *on the* $\mathfrak{l}$*-isotypic line of* X,

$$c_Q(\xi_+) + c_Q(\xi_-) - p_1 = f_O - f_{QO} - f_{\bar{Q}O} + f_{\partial O} \qquad (C5),$$

so the value $\frac{27}{4} - 8L$ *of [M, Cor c12-F0inv (iii)] at* $\psi = 0$ *is the value of a coboundary.*

Proof. Theorem 4.7. Dimensions: $\dim[p,q,r] = \frac{(p+1)(q+1)(r+1)(p+q+2)(q+r+2)(p+q+r+3)}{12}$: $[1,2,0] = 60$, $[0,1,1] = 20$. c_Q, p_1 as coefficients of $[\rho_0(\xi), \mathsf{e}]O$, $[\rho_0(P_b), \mathsf{e}]O$. □

Theorem 4.9 (Joining and lifting)**.** *Let* $E \subset \mathcal{V}$ *be a* ρ_0*-direct summand, a non-split extension* $0 \to N \to E \to C \to 0$ *of simple modules of the type of Theorem 4.7,* $C \ncong N$, *with* $N_w = 0$ *for* $w \le w_0$ *(*w_0 *the weight of* $C_{\rm top}$*). In an* $\mathfrak{l}$*-equivariant weight-preserving splitting* $E = C \oplus N$,

$$\rho_0 = \begin{pmatrix} \rho_C & 0 \\ \beta & \rho_N \end{pmatrix}, \qquad \delta = \begin{pmatrix} \delta_{CC} & \kappa \\ \delta_{NC} & \delta_{NN} \end{pmatrix}, \qquad \beta \in Z^1(\mathfrak{psl}(4|4), \mathfrak{l}; \mathrm{Hom}(C,N)),\ [\beta] \neq 0.$$

Under (H0), after the normalisation $\delta|_{\mathfrak{so}\oplus\mathfrak{so}} = 0$*:*

(i) $\delta(\Delta)|_E = -2\gamma\,\mathrm{Id}_E$ *(all submultiplets share* γ*).*

(ii) $\kappa \in Z^1(\mathfrak{psl}(4|4), \mathfrak{l}; \mathrm{Hom}(N,C))$*;* $[\kappa]$ *is invariant under every covariant coboundary, under* G_B *and the coupling; under the tree-level rescaling* $n \mapsto \mu n$ *of a basis of* N*:* $\beta \mapsto \mu^{-1}\beta$, $\kappa \mapsto \mu\kappa$, $\kappa\beta$ *fixed.*

(iii) $d_\rho\delta_{CC} = \kappa \smile \beta$, $(\kappa \smile \beta)(x,y) := \kappa(x)\beta(y) - (-1)^{|x||y|}\kappa(y)\beta(x)$, $(d_\rho f)(x,y) := f([x,y]) - [f(x), \rho_C(y)] - [\rho_C(x), f(y)]$. *With* $\tilde{c}(\Delta) := \mathrm{Id}_C$, $\tilde{c}|_{\mathfrak{so}\oplus\mathfrak{so}\oplus\mathfrak{n}_\pm} := 0$ *and* $\mathrm{tg}_C := [d_\rho\tilde{c}] \in H^2(\mathfrak{psl}(4|4), \mathfrak{l}; \mathrm{End}C)$*:*

$$-2\gamma\ \mathrm{tg}_C = [\kappa \smile \beta] \quad \text{in } H^2(\mathfrak{psl}(4|4), \mathfrak{l}; \mathrm{End}C).$$

(iv) For $\xi \in \Sigma$, $\eta \in \Sigma^*$, *a highest vector* $c_0 \in C_{\rm top}$ *with* $\rho_C(\xi)c_0 = 0$, $n_0 := \beta(\xi)c_0 \neq 0$, *the* $\mathfrak{l}$*-type of* $\xi \otimes c_0$ *absent from* $C_{w_0+1/2}$*:*

$$\kappa(\eta)\,\beta(\xi)\,c_0 = -\gamma\,(\xi,\eta)\,c_0.$$

(v) General direction ($\rho_C(\xi)c_0$ *possibly* $\neq 0$*,* $\mathfrak{l}$*-types of* $\xi \otimes c_0$ *in* $C_{w_0+1/2}$ *and of* c_0 *in* C_{w_0} *of multiplicity one): with* $\rho_0([\xi,\eta])c_0 = \mu c_0$, $\delta_{CC}(\xi)c_0 = a_\xi\,\rho_C(\xi)c_0$, $\delta_{CC}(\eta)\rho_C(\xi)c_0 = a'_\eta\,\mu\,c_0$,

$$(a_\xi + a'_\eta)\,\mu\,c_0 + \kappa(\eta)\beta(\xi)c_0 = -\gamma\,(\xi,\eta)\,c_0;$$

under a covariant coboundary $a_\xi \mapsto a_\xi + f_{w_0} - f_{w_0+1/2}$, $a'_\eta \mapsto a'_\eta + f_{w_0+1/2} - f_{w_0}$*:* $a_\xi + a'_\eta$ *is invariant. (iv) is* $\mu = 0$.

Proof. (i) $\mathrm{End}_{\mathfrak{psl}(4|4)}(E) = \mathbb{C}$: $\mathrm{Hom}(N,C) = \mathrm{Hom}(C,N) = 0$ and non-splitness; (R7). (ii) The CN block of ρ_0 vanishes, so the CN block of the cocycle identity contains ρ_C, ρ_N, κ only; $\kappa|_{\mathfrak{l}} = 0$. Covariant e: $\kappa \mapsto \kappa + \rho_C\mathsf{e}_{CN} - \mathsf{e}_{CN}\rho_N$; $\rho_0 \circ B$ has zero CN block. (iii) CC block: $(\delta\rho_0)_{CC} = \delta_{CC}\rho_C + \kappa\beta$, $(\rho_0\delta)_{CC} = \rho_C\delta_{CC}$. Write $\delta_{CC} = -2\gamma\tilde{c} + \delta'$, δ' relative; $d_\rho\tilde{c}$ is relative. (iv) The CC block of the identity at (ξ,η) on c_0: $\delta_{CC}([\xi,\eta])c_0 = \frac{1}{2}(\xi,\eta)(-2\gamma)c_0$; on the right $\rho_C(\eta)c_0 = 0$, $\beta(\eta)c_0 \in N_{w_0-1/2} = 0$, $\delta_{CC}(\eta)c_0 \in C_{w_0-1/2} = 0$, $\rho_C(\xi)c_0 = 0$, $\delta_{CC}(\xi)c_0 = 0$ (multiplicity condition); remaining $\kappa(\eta)\beta(\xi)c_0$. (v) The same identity with $\rho_C(\eta)\delta_{CC}(\xi)c_0 = a_\xi\rho_C(\eta)\rho_C(\xi)c_0 = a_\xi\mu c_0$ ($\rho_C(\eta)c_0 = 0$) and $\delta_{CC}(\eta)\rho_C(\xi)c_0 = a'_\eta\mu c_0$; coboundary: $[\rho_C(\xi), \mathsf{e}]c_0 = (f_{w_0} - f_{w_0+1/2})\rho_C(\xi)c_0$, $[\rho_C(\eta), \mathsf{e}]\rho_C(\xi)c_0 = (f_{w_0+1/2} - f_{w_0})\mu c_0$. □

Corollary 4.10 (Norms)**.** *Under the hypotheses of [M, Thm c8-adjoint] and of Theorem 4.9 (iv) with* $\eta = \bar{\xi}$*:*

$$\big\|\rho(\xi)c_0\big\|^2_{\rm rad} = (\xi,\bar{\xi})\,\gamma\,\|c_0\|^2_{\rm rad} + O(\hbar^2).$$

The one-loop norm of the classical null descendant is fixed by γ*; positivity requires* $\gamma \ge 0$ *(*$\gamma_K = \frac{3g^2N}{4\pi^2}$*).*

Proof. $\xi^\dagger = -\bar{\xi}$ [M, Thm c8-dag (i)]; $\langle\rho(\xi)c_0, \rho(\xi)c_0\rangle = -\langle c_0, \rho(\bar{\xi})\rho(\xi)c_0\rangle = -\langle c_0, \rho([\xi,\bar{\xi}])c_0\rangle$ ($\rho(\bar{\xi})c_0 = 0$) $= \big((\xi,\bar{\xi})(w_0+\gamma) - m\big)\|c_0\|^2$, m the eigenvalue of $\rho(M+R)$; order $\hbar^0$ on the null direction: $(\xi,\bar{\xi})w_0 = m$. □

Theorem 4.11 (Relations)**.** *Under (H0) the order-*$\hbar$ *parts of* $[D,J] = \lambda_J J$, $[Q,Q] = P$, $\{Q,S\} = D + M + R$, $[K,P] = D + M$, $\{S,S\} = K$ *and* $Q^2 = 0$ *(rank one) are the components (R7), (R2), (R3), (R5), (R6), (R4) of the cocycle identity [M, Cor c12-rel]. They hold in every scheme, gauge fixing and regularisation, exactly on* $\mathcal{V}$, *and on representatives modulo* $\Lambda^{(0)}(\chi_{x,y}(O))$ *[M, Thm c12-opalg (ii)], with* $\chi_{x,y}$ *transforming by the arity-two row of Theorem 3.1. Coboundaries and* $\rho_0 \circ B$ *satisfy them identically; a relation verified in one scheme is verified in all. The individual terms are scheme-dependent except* $\delta(\Delta)$.

Proof. [M, Thm c12-opalg (ii), (iii)]; $[\rho_0, \mathsf{e}]$ and $\rho_0 \circ B$ are cocycles. □

5 The twisted differential

Let $\xi = u \otimes z \in S_+ \otimes W$, $[\xi,\xi] = 0$, $Q^\xi = r_\xi(\Lambda)|_{q=1} = Q_0^\xi + \hbar Q_1^\xi + O(\hbar^2)$ [M, Lemma c8-ev, Def c8-twisted]:

$$\hbar Q_1^\xi = \hbar L_1^{(0)} + Q_1 - \tfrac{1}{2}\hbar L_1^{(2)}(\xi,\xi) - \tfrac{1}{6}\hbar L_1^{(3)}(\xi,\xi,\xi) - \cdots, \qquad E_1 := H(Q_0^\xi), \qquad d_1 := [Q_1^\xi] : E_1 \to E_1.$$

$N := \{S^{(1)}(R_\xi), -\}$, $[R_\xi, \xi] = \xi$; $N_t :=$ the operator multiplying a local functional of degree d and Weyl weight w by $t = d + 2w$.

Theorem 5.1 (Twisted class)**.**

(i) $d_1^2 = 0$.

(ii) Under $L_1 \mapsto L_1 + \partial\mathfrak{u}$: $Q_1^\xi \mapsto Q_1^\xi + [Q_0^\xi, r_\xi(\mathfrak{u})|_{q=1}]$, $r_\xi(\mathfrak{u})|_{q=1} = \mathfrak{u}^{(0)} + \mathfrak{u}^{(1)}(\xi) - \frac{1}{2}\mathfrak{u}^{(2)}(\xi,\xi) - \cdots$. *Under the coupling:* $-a(Q_0^\xi + [N, Q_0^\xi])$; *under* G_B: $+a[N, Q_0^\xi]$.

(iii) $Q_0^\xi = -[Q_0^\xi, N_t]$: *every multiple of* Q_0^ξ *is* $\mathrm{ad}_{Q_0^\xi}$*-exact in* End. *This is the precise sense of "*$\mathbb{C}\, S^\xi \subset \operatorname{Im} Q_0^\xi$*"* ($\mathrm{ad}_{S^\xi} = Q_0^\xi$); $S^\xi \in Q_0^\xi$(*local functionals*) *is a different statement, not implied (*N_t *is not of the form* ad_ϕ*:* $t(f) + t(f^*) = 7$*).*

(iv) Hence, under (Hs), (Hc): Q_1^ξ *is determined modulo* $\{[Q_0^\xi, h]\} + \mathbb{C}Q_0^\xi$, *and* $d_1 \in \mathrm{Hom}(E_1, E_1)$ *is invariant under (a)–(d), (f), including finite parts of every order and arity,* $\hbar F_1^{(0)}$, *the admissible* t, $\hbar F_1^{(2)}$, *the coupling,* G_B, *the profile, valid gauge fixings, and the homotopy data of a transfer to* $H(Q^\xi_{\mathrm{free}})$*; (e1) conjugates, (e2) and the rescaling* s_t *of [TW, Prop. Tree level] multiply* d_1 *by a constant. The higher differentials* d_r, $r \geq 2$, *are invariant likewise.*

(v) The arity-one Q_1 *alone is not invariant on* E_1: $\mathfrak{u}^{(1)}$ *changes it by* $\hbar[\Lambda_0^{(0)}, \mathfrak{u}^{(1)}(\xi)]$, *compensated in* Q_1^ξ *only together with* $-\frac{1}{2}\hbar[\Lambda_0^{(1)}, \mathfrak{u}^{(1)}]^{(2)}(\xi,\xi)$ *and* $[\Lambda_0^{(1)}(\xi), \mathfrak{u}^{(0)}]$.

(vi) On $H(\rho_0(\xi); \mathcal{V})$ *(gauge-invariant point operators),* $[\delta(\xi)]$ *is canonical:* $\{\rho_0(\xi), \delta(\xi)\} = 0$ *(R4),* $[\rho_0(\xi), \mathsf{e}]$ *and* $\rho_0(B\xi) = \rho_0(\xi)$ *induce* 0.

Proof. (i) Order $\hbar^2$ of $(Q^\xi)^2 = 0$ [M, Thm c8-twist]: $(Q_1^\xi)^2 + [Q_0^\xi, Q_2^\xi] = 0$. (ii) r_ξ is multiplicative and kills d_{CE} [M, Lemma c8-ev], so $r_\xi(U^{-1}LU) = r_\xi(U)^{-1} r_\xi(L) r_\xi(U)$; coupling and G_B: $r_\xi(S - S^{(2)})|_{q=1} = S^\xi + NS^\xi$, $r_\xi(G_B)|_{q=1} = -NS^\xi$, $\mathrm{ad}_{NS^\xi} = [N, Q_0^\xi]$ [M, Thm c8-twist proof]. (iii) Q_0^ξ raises t by 1: $\Lambda_0^{(0)}$: $(\Delta d, \Delta w) = (1, 0)$; $\{S^{(1)}(\xi), -\}$: $(0, \frac{1}{2})$; $\{S^{(2)}(\xi,\xi), -\}$: $(-1, 1)$. So $[N_t, Q_0^\xi] = Q_0^\xi$. ad_ϕ for $\phi = \sum y_f \int (f, f^*)$ multiplies f by y_f and f^* by $-y_f$; $t(f) + t(f^*) = -1 + 2 \cdot 4$. (iv) (ii), (iii), Theorems 2.5, 2.3, 2.4; a closed G with $G^{(0)} = G^{(1)} = 0$ is exact under (Hs); homotopy data: two transfers are related by an L_∞ quasi-isomorphism, $\tilde{Q}_1 \mapsto \tilde{Q}_1 + [\tilde{Q}_0, \phi_h]$; d_r: spectral sequence of the $\hbar$-filtration, invariant under filtered conjugations. (v) Theorem 3.1. (vi) Corollary 3.2, (R4). □

Theorem 5.2 (Pair entries)**.** *After transfer to the cohomology* $H(Q^\xi_{\mathrm{free}})$ *the letters are* $\mathsf{c}, \gamma, \beta, \mathsf{b}$ *and holomorphic derivatives [TW, Prop. Single-letter cohomology]. On these letters* $\tilde{Q}_0$ *raises the word length by one and adds no derivative (*$\tilde{Q}_0 C = \frac{1}{2}[C, C]$ *[BK, (3.49)–(3.50)], [TW, Prop. Tree level]). For* $\tilde{\mathfrak{u}}$ *write* $\tilde{\mathfrak{u}}_{n\to m}$ *for its component from* n*-letter to* m*-letter words. Then on two-letter words*

$$[\tilde{Q}_0, \tilde{\mathfrak{u}}]_{2\to 2} = \tilde{Q}_0\, \tilde{\mathfrak{u}}_{2\to 1} - \tilde{\mathfrak{u}}_{3\to 2}\, \tilde{Q}_0.$$

For underived letters X, Y *and the structure* $\langle U \cdot V \rangle = f_{ACD} f_{BCE}\, \varepsilon^{\dot\alpha\dot\beta} \partial_{\dot\alpha} U^D \partial_{\dot\beta} V^E$*:*

(i) $\tilde{Q}_0 \tilde{\mathfrak{u}}_{2\to 1}(XY)$ *has no component along* $\langle U \cdot V \rangle$*;*

(ii) $(\tilde{\mathfrak{u}}_{2\to 1} \otimes 1)\tilde{Q}_0(XY)$ *and all one-body terms keep an underived spectator, hence no component along* $\langle U \cdot V \rangle$*;*

(iii) only a genuinely three-letter $\tilde{\mathfrak{u}}_{3\to 2}$ *moves the coefficient of* $\langle U \cdot V \rangle$ *in* $\tilde{Q}_1(XY)$.

Field redefinitions, the coupling, G_B, *the admissible* t, *finite parts acting on at most two letters and their arity components, profile changes and the homotopy data have no genuinely three-letter part. Hence the reduced pair entries of [TW, Thm. Pair entries] are invariant in this class (no rotation covariance needed), are independent of the profile, and are rational in the heat-kernel scheme (Corollary 6.5): every* $\log \frac{3}{2}$ *cancels, as found in [TW]. A finite part on three letters moves them ([M, Rem c11-twist]:* $\psi(W) = y \operatorname{tr} \partial\mathsf{c}\, \partial\mathsf{c}$, *shift* $-3y$ *of the Konishi entry). For derived inputs (ii) fails (a spectator may carry the derivative).*

Proof. $\tilde{Q}_0$ is $n \to n+1$, so $(\tilde{Q}_0\tilde{\mathfrak{u}})_{2\to 2} = \tilde{Q}_0\tilde{\mathfrak{u}}_{2\to 1}$, $(\tilde{\mathfrak{u}}\tilde{Q}_0)_{2\to 2} = \tilde{\mathfrak{u}}_{3\to 2}\tilde{Q}_0$. (i) $\tilde{\mathfrak{u}}_{2\to 1}(XY)$ is one letter Z; a component along two first derivatives needs $Z = \partial_{\dot\alpha}\partial_{\dot\beta}Z'$, symmetric; $\tilde{Q}_0 Z = \partial_{\dot\alpha}\partial_{\dot\beta}\tilde{Q}_0 Z'$ contributes $\partial_{(\dot\alpha}U\partial_{\dot\beta)}V$ only, killed by $\varepsilon^{\dot\alpha\dot\beta}$. (ii) $\tilde{Q}_0(XY)$ is a sum of three underived letters. (iii) (i), (ii). Transfer: terms $p\,\mathfrak{u}\,h\,Q_{\mathrm{int}}\,i$, $p\,Q_{\mathrm{int}}\,h\,\mathfrak{u}\,i$ with Q_{int} $1\to 2$ and h one-body have no $3\to 2$ part beyond $(\mathfrak{u}_{2\to 1}\otimes 1)$; field redefinitions and N are one-body; Q_0^ξ is $n\to n+1$; the regularisation change $\psi_{RR'}$ comes from one-loop graphs, which meet an operator in at most two letters [M, R8]; profile: Theorem 2.3. □

Remark 5.3 (On [TW, Thms. Class, Entries])**.** (A) [TW, Thm. Class] (i)–(iii) holds with: free directions of $\hbar F_1$ = elements of Z^0 (all arities imposed); t_ξ alone and θ_V^ξ, $\theta_\perp^\xi$ separately are not free ([M, Thm c12-ar2]); admissible t act by $c[\rho_0,\rho_0(\Delta)]$ and $-a\rho_0\circ B$, which induce 0 on E_1 through r_ξ (Theorem 5.1 (ii)); $\hbar F_1^{(2)}$ directions are s-exact under (Hs); "$\mathbb{C}\,S^\xi \subset \operatorname{Im} Q_0^\xi$" in the operator sense of Theorem 5.1 (iii). Added: profiles and valid gauge fixings (Theorems 2.3, 2.4). (B) [TW, Thm. Entries] holds; Theorem 5.2 needs no rotation covariance, only the absence of genuinely three-letter components after transfer. The Konishi entry is carried by $\hbar L_1^{(0)}$ (arity-zero kernel with output ∂c) [TW, Rem. on Tab. c11-twistentries], which is Theorem 5.1 (v): the arity-one Q_1 alone misses it.

Proposition 5.4 (No analogue for generic Q, S, K)**.** *Let $x \in \{\xi\ (\Gamma(\xi,\xi)|_V \neq 0),\ \eta,\ K_b\}$. $\rho_0(x)$ has a word-length-preserving part $\rho_0(x)_{1\to 1}$ on letters (the free transformation), so for a two-body finite part $[\rho_0(x),\mathsf{e}]_{2\to 2} = [\rho_0(x)_{1\to 1},\mathsf{e}_{2\to 2}] + \cdots \neq 0$ in general: e.g. $\mathsf{e} = f_O$ on $O = \int \omega(\iota_{w_1}A,\iota_{w_2}A)$ and $f_{\mathcal{B}}$ on $\mathcal{B}_1 = \{S^{(1)}(\xi),O\}$ shifts c_Q by $f_O - f_{\mathcal{B}}$. No Q_0-cohomology exists ($\rho_0(\xi)^2 = \frac{1}{2}\rho_0([\xi,\xi]) \neq 0$, a translation). The two-body kernels of Q_1, S_1, K_1 are not invariant even in one-loop-local schemes; their invariant content is Theorems 4.5–4.11. Along Δ, $[\rho_0(\Delta),\mathsf{e}] = 0$: D_1 with its kernels is invariant (Theorem 4.1).*

6 Profiles

Definition 6.1 (Profiles)**.** A profile is a smooth $\varpi : [0,\infty) \to \mathbb{R}$, $\varpi(0) = 1$, rapidly decreasing; $P_\varpi(\varepsilon,T) := (Q^{GF}\otimes 1)\big((\varpi(\varepsilon H) - \varpi(TH))H^{-1}\otimes 1\big)K_0$, $K_t^\varpi := (\varpi(tH)\otimes 1)K_0$, special edge $K_\varepsilon^\varpi - K_0$. Heat kernel: $\varpi_1(\mathfrak{u}) = e^{-\mathfrak{u}}$. Family $\varpi_k(\mathfrak{u}) := e^{-\mathfrak{u}^k}$, $k\in\mathbb{Z}_{\geq 1}$. $\vartheta := \mathfrak{u}\frac{d}{d\mathfrak{u}}$; $c_E[\varpi] := \int_0^\infty \frac{d\mathfrak{u}}{\mathfrak{u}}\varpi^E(\varpi - 1)$.

Lemma 6.2. *$(Q\otimes 1 + 1\otimes Q)P_\varpi(\varepsilon,T) = K_\varepsilon^\varpi - K_T^\varpi$; [M, Lemma c8-transfer] and the constructions of [M, ch. 8, 12] hold with K^ϖ in place of K.*

Proof. $[Q,Q^{GF}] = H$, Q commutes with functions of H; the proof of [M, Lemma c8-transfer] uses only $[\partial_\phi, Q] = \partial_{(Q\otimes 1+1\otimes Q)\phi}$. □

Lemma 6.3 (Mellin form)**.** *For a one-loop defect cycle (special edge, edges P_ϖ, possibly one composite line, injected momentum r) of superficial degree D, the coefficient of ε^0 is*

$$a_0 = \frac{1}{16\pi^2}\int_0^\infty \frac{d\mathfrak{u}}{\mathfrak{u}}\,\Phi(\mathfrak{u}), \qquad \Phi \in \mathbb{Q}(i)[p,r]_D \otimes \mathbb{Q}[\varpi,\vartheta\varpi,\vartheta^2\varpi,\dots],$$

the coefficients independent of ϖ, each monomial containing the special-edge factor $\varpi - 1$ or $\vartheta^m\varpi$, $m\geq 1$.

Proof. $l = l'/\sqrt{\varepsilon}$: ε enters only through $\varpi\big((l' + \sqrt{\varepsilon}q)^2\big)$; the order-$D$ Taylor term in $\sqrt{\varepsilon}q$ is homogeneous of degree 0; $(\mathfrak{u}^m\varpi^{(m)})(\mathfrak{u}) = \prod_{i=0}^{m-1}(\vartheta - i)\varpi$; angular averages rational; $\int d^4l' = \pi^2\int \mathfrak{u}\,d\mathfrak{u}$; the special-edge factor vanishes at $\mathfrak{u} = 0$, rapid decay at ∞; the remainder is $o(1)$ by dominated convergence. □

Theorem 6.4 (Profile dependence)**.** *For a monomial $\mathcal{M}$ with $m(\mathcal{M})$ ϑ's,*

$$\int_0^\infty \frac{d\mathfrak{u}}{\mathfrak{u}}\mathcal{M}[\varpi_k] = k^{m(\mathcal{M})-1}\int_0^\infty \frac{d\mathfrak{u}}{\mathfrak{u}}\mathcal{M}[\varpi_1], \qquad \int \frac{d\mathfrak{u}}{\mathfrak{u}}\mathcal{M}[\varpi_1] \in \begin{cases} \sum_j \mathbb{Q}\log j, & m = 0,\\ \mathbb{Q}, & m\geq 1.\end{cases}$$

Hence every coefficient X obtained from the a_0's by a ϖ-independent $\mathbb{Q}(i)$-linear map (obstructions, $\hbar F_1$ in a fixed basis with fixed reference parts, kernels Υ_x, one-body terms, components on fixed representatives) satisfies

$$X(k) = \frac{\Lambda_X}{k} + \sum_{j\geq 0} k^j R_{X,j}, \qquad \Lambda_X \in \textstyle\sum_j \mathbb{Q}(i)\log j \quad (\textit{the log part of } X(1)), \quad R_{X,j}\in\mathbb{Q}(i).$$

Proof. $\vartheta(f\circ \mathfrak{u}^k) = k(\vartheta f)\circ\mathfrak{u}^k$, $\frac{d\mathfrak{u}}{\mathfrak{u}} = \frac{1}{k}\frac{d\mathfrak{v}}{\mathfrak{v}}$. $m = 0$: $\int \frac{d\mathfrak{u}}{\mathfrak{u}}P(e^{-\mathfrak{u}})$, $P(1) = 0$, Frullani: $-\sum p_j\log j$. $m\geq 1$: $\vartheta^m e^{-\mathfrak{u}} = \mathfrak{u}\,q_m(\mathfrak{u})e^{-\mathfrak{u}}$, $\int \mathfrak{u}^{a-1}e^{-b\mathfrak{u}}d\mathfrak{u} = \Gamma(a)b^{-a}\in\mathbb{Q}$. $1,\log 2,\log 3,\dots,\log p$ (p prime) are $\mathbb{Q}(i)$-independent (Lindemann–Weierstrass and unique factorisation). □

Corollary 6.5 (Rationality criterion)**.** *(i) If $\Lambda_X \neq 0$, X depends on the profile: $X(2)-X(1) = -\frac{1}{2}\Lambda_X + \mathbb{Q}(i) \neq 0$. (ii) If X is profile-independent, then $\Lambda_X = 0$, $R_{X,j} = 0$ $(j \geq 1)$, $X(1) \in \mathbb{Q}(i)$, and X equals its single-ϑ part, which is universal: $\int \frac{d\mathsf{u}}{\mathsf{u}}(\vartheta\varpi)G(\varpi) = \int_1^0 G(y)\,dy$ for every profile.*

Corollary 6.6 (The claimed combinations)**.** *$t_{\xi_+} + t_{\xi_-} - t_P = \frac{27}{4} + 2L$, $t_{\xi_+} + t_{\eta_+} - t_\Delta = \frac{27}{4} + 2L$ [M, Thm c12-ar2], $c_Q(\xi_+) + c_Q(\xi_-) - p_1 = c_Q(\xi_+) + c_S(\eta_+) - \frac{1}{4}D_1|_{\text{traceless}} = \frac{27}{4} - 8L$ [M, Cor c12-F0inv] (C3), $c_Q = \frac{79}{12} - 3L - t_\xi = -2L - \theta_\perp - p_\perp$ (C4), $\hbar\mathrm{Ob}^{(1)}(\xi)|_2 \ni (\frac{79}{24} - \frac{1}{2}L)$, every coefficient with a $\log\frac{3}{2}$ part: profile-dependent. Under ϖ_k the log parts become $\frac{2}{k}L$, $\frac{2}{k}L$, $-\frac{8}{k}L$, $-\frac{3}{k}L$, $-\frac{1}{2k}L$. Rational numbers ($t_\Delta = -\frac{1}{3}$, $\frac{27}{16}$, $-\frac{5}{2}$) are not decided by Theorem 6.4.*

Example 6.7 (Second profile, C7)**.**

$$J_0(q) = \frac{4}{\pi}\int_0^\infty l^3 dl \int_0^\pi \sin^2\nu\, d\nu\, \frac{(\varpi(l^2)-1)\,\varpi(|l+q|^2)}{l^2\,|l+q|^2} \xrightarrow{q\to 0} c_1[\varpi],$$

$$J_2(q) := \big(J_0 \text{ without the factor } l^{-2}\big) = A + Bq^2 + O(q^4),$$

$$A = \int_0^\infty d\mathsf{u}\,(\varpi-1)\varpi, \qquad B = \tfrac{1}{2}\int_0^\infty d\mathsf{u}\,\mathsf{u}\,\varpi''(\mathsf{u})\,(\varpi(\mathsf{u})-1)\,:$$

	$c_1[\varpi]$	A	B	B (general k)
ϖ_1	$-\log 2$	$-\frac{1}{2}$	$-\frac{3}{8}$	$-\frac{k}{8} - \frac{1}{4}$
ϖ_2	$-\frac{1}{2}\log 2$	$-\frac{\sqrt{\pi}}{2}\big(1 - \frac{1}{\sqrt{2}}\big)$	$-\frac{1}{2}$	
numerics	-0.6925, -0.3474 $(q = 0.05)$	-0.5000003, -0.2595694	-0.37495, -0.50008	

$D = 0$: log part $\propto 1/k$; $D = 2$: rational part linear in k.

7 Field countings and the action-side relations

Proposition 7.1 (Field countings)**.** *For $\phi_y = \sum_f y_f \mathcal{N}_f$ $(y_{A|_V} = y_{A|_{W_{\mathbb{R}}^\perp}} = y_A)$, $\hbar F_1 + a\hbar\, s\phi_y$ is a one-loop quantisation for all a. With $\hbar F_1^{(1)}(\xi)|_2 = p(T_1+T_2) + t_\xi(T_1+T_2+T_3+T_4)$ [M, Thm c12-sol] and the arrow rule of [M, Prop c11-freedom (i)] (T_1, T_2: $A \to \lambda$; T_3, T_4: $\lambda \to A$):*

$$t_{\xi_\pm} \mapsto t_{\xi_\pm} + \sigma a(y_A - y_\lambda), \qquad p \mapsto p - 2\sigma a(y_A - y_\lambda), \qquad t_P,\ t_\Delta,\ p_1,\ D_1 \text{ unchanged},$$

$$c_Q(\xi_\pm) \mapsto c_Q(\xi_\pm) + \sigma a(y_A - y_\lambda), \qquad (t_{\xi_+} + t_{\xi_-} - t_P),\ (c_Q(\xi_+) + c_Q(\xi_-) - p_1) \ \mapsto\ +2\sigma a(y_A - y_\lambda),$$

$\sigma = \pm 1$ the bracket sign (C2). Both combinations take every value on the solution space at fixed regulator, gauge fixing and $\psi = 0$.

Proof. $s\phi_y \in B^0$. $\{\phi_y, S^{(1)}(\xi)\}$ multiplies the arrow $f \to f'$ by $\pm(y_{f'} - y_f)$; $S^{(1)}(P_b)$, $S^{(1)}(\Delta)$ preserve the species. $c_Q = -2L - $ (coefficient of T_2) [M, Thm c12-F0 proof]. C2. □

Corollary 7.2 (What the action-side relations are)**.** *The relations [M, eq. (c12-t), Cor c12-inv] are invariant exactly under the subgroup generated by the inner direction, G_B, the coupling and the admissible t, at fixed $\hbar F_1^{(0)}|_2 = \frac{27}{16}\int(d^*A|_V, d^*A|_V)$ units, fixed reference particular parts and fixed regularisation; [M, eq. (c12-F0inv)] in addition at $\psi = 0$. They move under field countings (Proposition 7.1), finite parts (C5), profiles (Corollary 6.6) and gauge fixing. Their invariant content: solvability of the arity-two equations ($\varrho = \partial$ of a diagonal cochain [M, Thm c12-fix]); on $V_{20'}$, $[\delta] = 0$ (Corollary 4.8).*

8 Answer: defined up to cohomology?

Theorem 8.1. *Under (H0), (Hs), (Hc), at one loop:*

generator	representative is defined modulo	canonical content
twisted Q (ξ rank one)	$[Q_0^\xi, h] + \mathbb{C}Q_0^\xi$, h local, for the combination $Q_1^\xi = L_1^{(0)} + L_1^{(1)}(\xi) - \frac{1}{2}L_1^{(2)}(\xi,\xi) + \cdots$	$d_1 \in \mathrm{Hom}(E_1, E_1)$, its kernel, image, E_2; pair entries on underived letters in one-loop-local schemes; rational. "Up to cohomology": yes, exactly.
Q along rank-one ξ, arity one alone	$\hbar[\Lambda_0^{(0)}, \mathfrak{u}^{(1)}(\xi)] + \hbar[\Lambda_0^{(1)}(\xi), \mathfrak{u}^{(0)}]$, $-a\rho_0(\xi)$	$[\delta(\xi)]$ on $H(\rho_0(\xi); \mathcal{V})$ only
generic Q, S	chain: homotopy $[\Lambda_0^{(0)}, \mathfrak{u}^{(1)}(x)]$; on $\mathcal{V}$: $[\rho_0(x), \mathsf{e}]$ with one e for all generators, $\mp a\rho_0(x)$	no Q_0-cohomology; content in $[\delta]$: lifting classes $[\kappa]$, $\kappa(\eta)\beta(\xi) = -\gamma(\xi,\eta)$, norms, (R2), (R3)
K	as S; finite T, T-independent	(R5) with D_1; $[\kappa]$ components along K
D	chain: homotopy; on $\mathcal{V}$: nothing	$\delta(\Delta)$ itself: matrix, spectrum, eigenvectors; position part $2x_O \cdot P_1$ is scheme
P	$[\rho_0(P_b), \mathsf{e}]$	none on simple summands; $P_1 = 0$ is a renormalisation condition
M, R	$[\rho_0(x), \mathsf{e}]$	none (Whitehead): pure gauge

Proof. Theorems 3.1, 4.1, 4.5, 4.7, 4.9, 5.1, 5.2, Corollary 4.10, Proposition 5.4; P, M, R: [M, Cor c12-rel (R8)] and Theorem 4.7 (ii). □

9 The computed numbers

Status: [I] invariant (theorem), [A] artefact (moved by the freedom named), [C] coordinate (fixed by a convention), all from the old gauge fixing (Remark 11.1).

quantity	status	reason
$D_1 = 0$ (20 traceless), -48 units (trace) $= \frac{3g^2N}{2\pi^2} = 2\gamma_K$	[I]	Thm 4.1; rational (Cor 6.5); C9
spectra: $-8\sum(1-P)$, $-8\sum(1-\Pi)$, $-32h(j)$	[I]	Thm 4.1; rational; = [B03c]
$t_\Delta = -\frac{1}{3}$, θ_d^Δ, two-body/one-body split of D_1	[C]	gauge fixing: composite lines along Δ cancel for the valid Q^{GF} [GF, F3]
$t_{\xi_+} + t_{\xi_-} - t_P = t_{\xi_+} + t_{\eta_+} - t_\Delta = \frac{27}{4} + 2L$	[C]	Prop 7.1, Cor 6.6
$c_Q = \frac{79}{12} - 3L - t_\xi$, $c_S = \frac{79}{12} - 3L - t_\eta$	[A]	coboundary, G_B, field counting, profile
$c_Q(\xi_+) + c_Q(\xi_-) - p_1 = \frac{27}{4} - 8L$	[A]	value of a coboundary on $V_{20'}$ (Cor 4.8); profile; field counting
$4L(\mathcal{B}_2 + \mathcal{B}_3)$ in Q_1 (trace)	[A]	two-body kernel (Prop 5.4); profile
$\hbar L_1^{(0)}(O) =$ $-(2+4L)(w_1,w_2)\int\omega(dc, A\|_V)$	[A]	removed by ψ [M, Cor c12-F0inv (iv)]
$P_1 = -(\frac{1}{3} + t_P)\partial_b$, $M_1 = -(\frac{1}{3} + t_M)\rho_0(M)$, $R_1 = 0$	[A]	pure gauge (Thm 8.1)
K_1 one-body $\frac{44}{3}$, θ_d^K, κ_j^K	[C]/[A]	position part of D_1^0; open directions
$\hbar F_1^{(1)}(\eta) \ni -\frac{5}{2}u_0$, $u_0 = \int(\rho(A\|_V)\eta, \lambda^*)$	[C]	gauge sector: pairs with $\int(\eta,\lambda)d^*A\|_V$, $\int(\rho(dc)\eta,\lambda^*)$; rational, not decided by Thm 6.4; same value $-\frac{5}{2}$ on $V\lambda^\ddagger\bar{\alpha}^\#$ at gauge parameter 1 in the earlier N=4 project (survey): consistent with a common Feynman-gauge origin, no invariant meaning
$\hbar\mathrm{Ob}^{(1)}(\xi)\|_2$, $\frac{27}{16}$, $\hbar F_1^{(1)}(\xi)\|_2$ particular	[C]	regulator representative (Thm 2.3)
$L^{(1)}(\xi)\|_2 = 2(T_1 + T_2 - T_3 - T_4)$ units	[C]	log coefficient: profile-universal, gauge-fixing dependent; $= \{\mathcal{N}, S_2^{(1)}(\xi)\}$ units, generator of the ε-unit freedom
$Q_1^\xi = 0$ on chiral bilinears [M, Prop c12-twist]	[I]	class level (Thm 5.1); [BK, (3.77)] $n = 2$
twisted pair entries [TW, Thm. Pair entries]: $-\frac{i}{4}\mathcal{N}Q_1^{BK}$ in 8 channels	[I]	in one-loop-local schemes (Thm 5.2); rational

arity-one entries of [M, Tab c11-twistentries] ($\frac{64}{3} - 128L$, $32 + 64L$, ...)	[A]	arity one alone (Thm 5.1 (v)); Konishi entry needs the arity-zero kernel [TW, Rem. on Tab. c11-twistentries]
SQED: $-\frac{1}{12} - \frac{3}{2}\log 2 + \frac{3}{2}\log 3$ [SQ]	[A]	composite finite part $(c\phi_+)^{\mathrm{fin}}_{\mathrm{CT}}$ = (b)/(f)

10 Beisert and Budzik–Kulp

Proposition 10.1 (Dictionary with [B03d], [B03c])**.**

- *Similarity transformations $J(g) \mapsto T(g)J(g)T(g)^{-1}$ [B03d, (4.4)]: the gauge group $1 + \hbar\mathfrak{u}$ (b), (f), (a1). Tree level $e^{2i\beta_1 H_0 + i\beta_2 L}$ [B03d, (3.39)]: $\beta_1 \leftrightarrow$ inner direction ($\mathsf{e} \propto \rho_0(\Delta)$); β_2 (length) $\leftrightarrow$ (e3). Order g^2: T_2 from the structures of H_2, $\gamma_{1..4}$ [B03d, §4.2] $\leftrightarrow \mathsf{e} \in \mathrm{End}(\mathcal{V})_0$.*
- *Prediagonalisation $[H_0, H(g)] = 0$ [B03d, (3.18)] $\leftrightarrow$ weight-preserving schemes (homogeneity of δ).*
- *"Gauge transformations" (structures vanishing on cyclic states, $\delta_{1,2}$ [B03d, (3.6)]) $\leftrightarrow \Lambda^{(0)}$-exact terms and homotopies.*
- *Coupling redefinition $g \mapsto \alpha_1 g + \alpha_3 g^3$, $H_4 \mapsto \alpha_1^4 H_4 + 2\alpha_1\alpha_3 H_2$ [B03d, (4.9)–(4.10)] $\leftrightarrow S - S^{(2)}$ (invisible on δ at one loop; at two loops shifts $\delta_2(\Delta)$ by a multiple of $\delta(\Delta)$) and (e2).*
- *$[J_0, D_2] = 0$ [B03c, (2.10)], [B03d, (3.22)] $\leftrightarrow$ (R7), Theorem 4.1; normalisation constants irrelevant at one loop [B03c, App. B] $\leftrightarrow$ Theorem 4.1.*
- *Equal energy shifts of submultiplets at $O(g^{2\ell})$, joining at $O(g^{2\ell+1})$ [B03d, §3.7], Konishi submultiplets $E = 4C_2 = 6C_1 = 6C_1 - 2C_0$ [B03c, (6.1)] $\leftrightarrow$ Theorem 4.9 (i).*
- *$c_9 c_{10} = \frac{1}{2}c_1$ [B03d, (3.38)], $c_9 \propto \alpha_1 e^{i(\beta_1+\beta_2)}$, $c_{10} \propto \alpha_1 e^{-i(\beta_1+\beta_2)}$ (C9): the product of the normalised Q_1, S_1 is fixed by $H_2 \leftrightarrow$ Theorem 4.9 (iv), Corollary 4.10. In the BV normalisation β is classical ($O(1)$) and κ one-loop; in Beisert's normalisation both are $O(g)$.*
- *Hypercharge B_0, conserved by D_2 [B03c, after (2.10)] $\leftrightarrow$ Proposition 4.3.*

Remark 10.2 ([BK] and [HB])**.** [BK] define Q_1 on $H(Q_0)$ of the holomorphic theory with relations [BK, (3.59)–(3.63)] for the commutant $\mathfrak{psl}(3|3)$; these are (R)-relations at (ξ, R) restricted by r_ξ [M, Thm c11-twistcheck (d)]. Theorem 5.1 gives the invariance stated as open in [HB, Rem scheme]: at one loop d_1 is canonical in the equivariant construction, under every freedom of Theorem 2.1. The comparison $\Phi \circ d_1 \circ \Phi^{-1} = \mathrm{const} \cdot [Q_1^{BK}]$ [TW, Thm. Pair entries] is a comparison of canonical objects; the constant is the normalisation of $\hbar_{\mathrm{tw}}$ (s_t). The cancellation of all $\log\frac{3}{2}$ is forced by Theorem 5.2 and Corollary 6.5.

11 Gauge fixing of the data; open

Remark 11.1 (Old gauge fixing)**.** The data of [M] use a Q^{GF} with $(Q^{GF})^2 \neq 0$ and $\mathrm{Im}\, Q \cap \mathrm{Im}\, Q^{GF} \neq 0$, outside [C07, Def 4.3.3] (3), (5). Theorems 3.1–4.11, 5.1, 5.2, 6.4 use only an L with $L \circ L = 0$ and the transfer lemma, and apply to the old data as data. Theorem 2.4 does not apply to the old Q^{GF}: no theorem identifies old invariants with those of the valid gauge fixing of [GF] (frame Ψ, pair $(\bar{c}, \bar{c}^*)$, propagator AA, $\lambda\lambda$, $c \otimes \bar{c}$, $\bar{c} \otimes c$; arity-zero theory identical through the chain map j [GF, F2]). Survives as prediction: the [I] rows (to be recomputed; spectra already equal [B03c]); the twisted class (rational, equal to [BK]). Not inferable: every [C], [A] row, in particular every number with a $\log\frac{3}{2}$ part (Corollary 6.6) and $t_\Delta = -\frac{1}{3}$ (composite lines along Δ, P, M, R cancel for the valid Q^{GF} [GF, F3]).

Remark 11.2 (Open)**.** (1) The Konishi block: κ, β and Theorem 4.9 (iv) on the trace bilinear (needs S_1 on weights $\frac{5}{2}, 3$ and the arity-two kernels); Corollary 4.10 with Gram_1. (2) $V_{20'}$ a simple direct summand of the classical interacting module (hypothesis of Corollary 4.8). (3) (Hs) at three and four legs; G_j^K. (4) (Hc) local representatives in families. (5) DR inside BV: Slavnov–Taylor identity with operator insertions, so that Theorem 2.3 applies. (6) Twisted entries with derived inputs; normalisation of $\hbar_{\mathrm{tw}}$. (7) Two-loop δ_2 for Theorem 4.6. (8) All [I] rows in the valid gauge fixing.